Training Compass

The Smol Training Playbook

1. Why are you training this model?
2. What model should you train?

WHY?

We have compute available

↳ *That's a resource, not a goal*

Everyone else is doing it

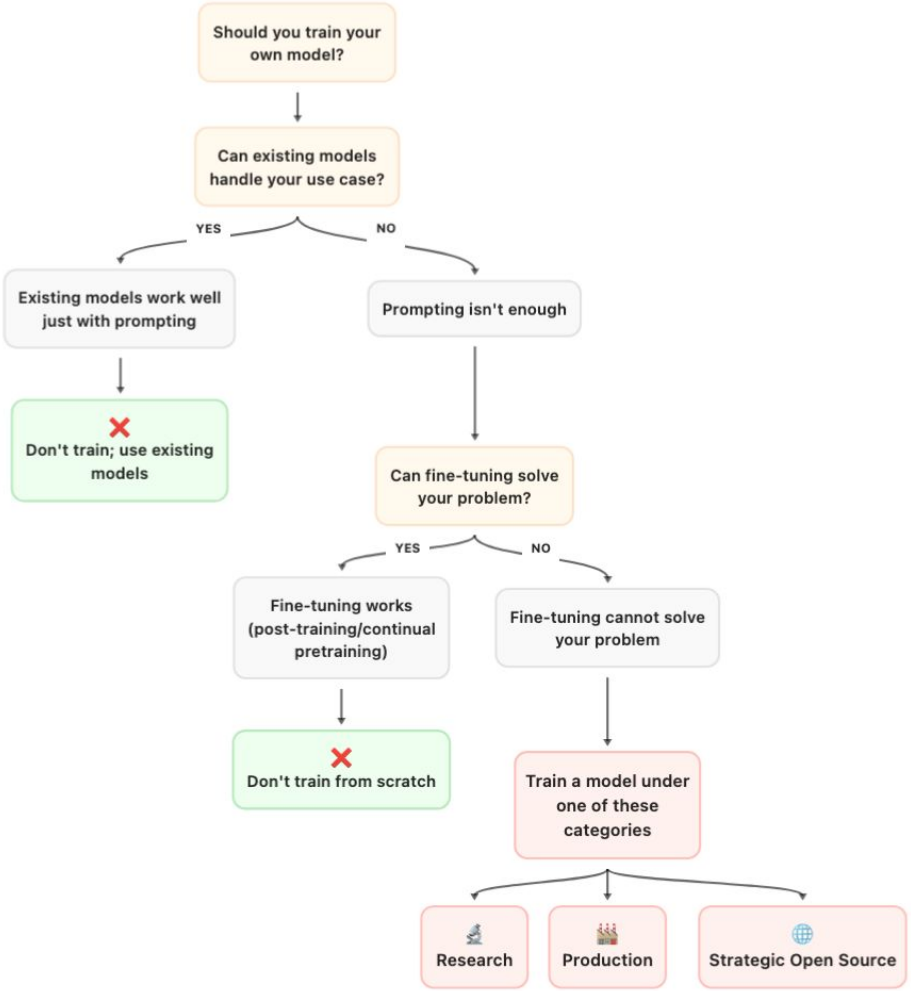
↳ *That's peer pressure, not strategy*

AI is the future

↳ *That's a platitude, not a plan*

We want the best model possible

↳ *That's not specific enough to guide decisions*


Should you train your own model?
Can existing models handle your use case?
YES
NO
Existing models work well just with prompting
Prompting isn't enough
Don't train; use existing models
Can fine-tuning solve your problem?
YES
NO
Fine-tuning works (post-training/continual pretraining)
Fine-tuning cannot solve your problem
Don't train from scratch
Train a model under one of these categories
Research
Production
Strategic Open Source

RESEARCH

- Can we scale training on this new optimizer to a 10B+ model?
- Can reinforcement learning alone, without supervised fine-tuning (SFT), produce reasoning capabilities?
- Can we train good small models on purely synthetic textbook data?
- Can we achieve competitive performance by training on only openly licensed data?

PRODUCTION

- Domain specificity: DNA model, legal or financial model
- Deployment constraints: resource constraints, custom hardware like FPGA
- Safety and governance: complete control over data and model behavior

STRATEGIC OPEN-SOURCE

~~STRATEGIC~~

OPEN-SOURCE

- Infrastructure for others
- Technical credibility

HF MOTIVATION

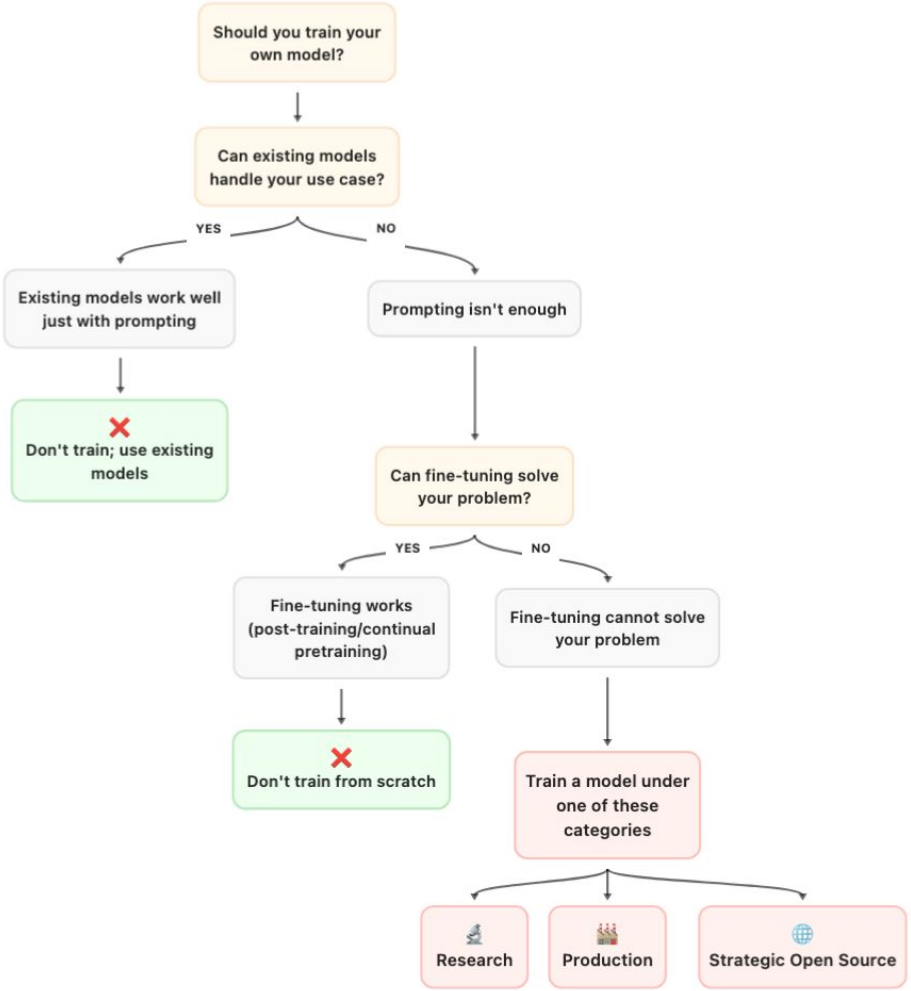
- TL;DR: concrete gaps in OSS eco-system
- BLOOM - OSS alternative to GPT-3
- StarCoder - OSS version of Codex
- SmolLM - few strong small models
- Zephyr - DPO works at scale
- Open-R1 - reproduce Deepseek R1

WHAT?

- Fast model for on-device → small efficient model
- Multilingual model → large tokenizer vocab
- Super long context → hybrid architecture

HOW?

- Iteration speed
- Data curation: 💩 → 💩
- Team size: start small, reeeeeally small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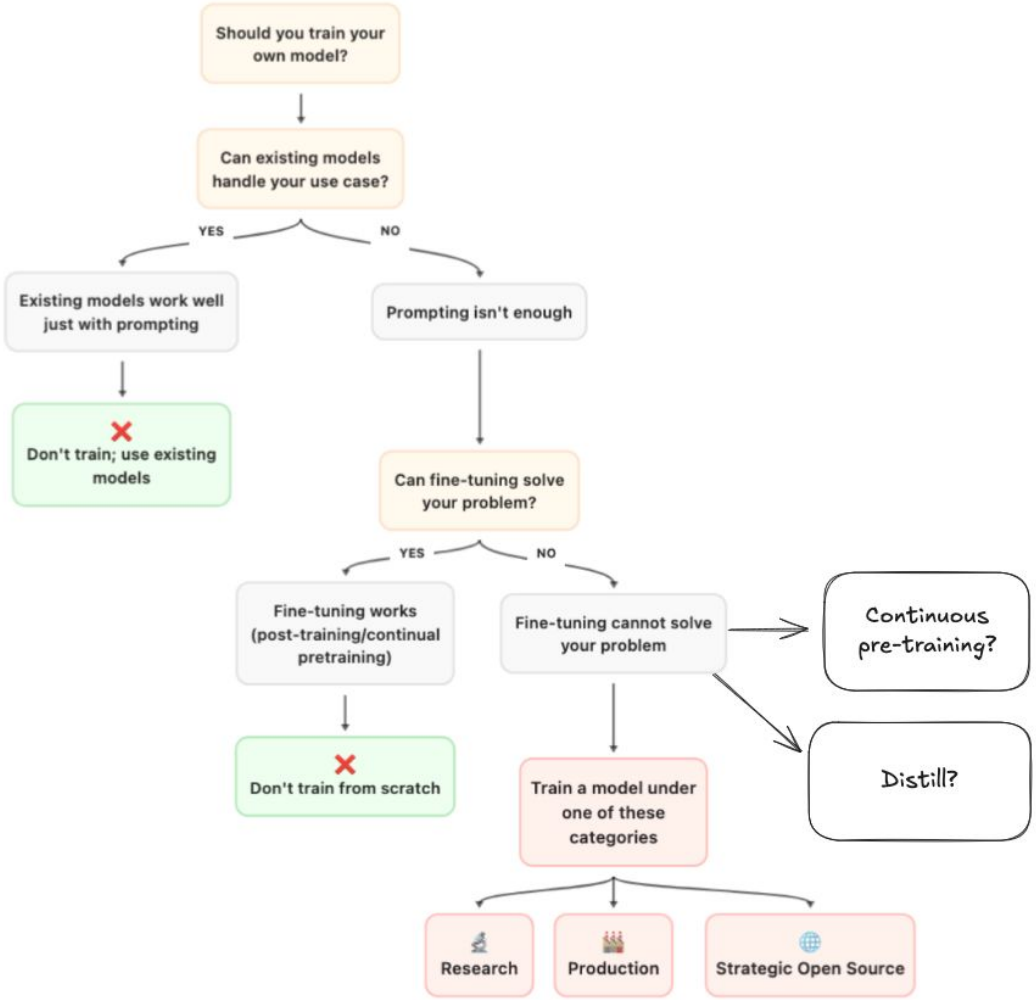
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Continuous pre-training?
Distill?
Don't train from scratch
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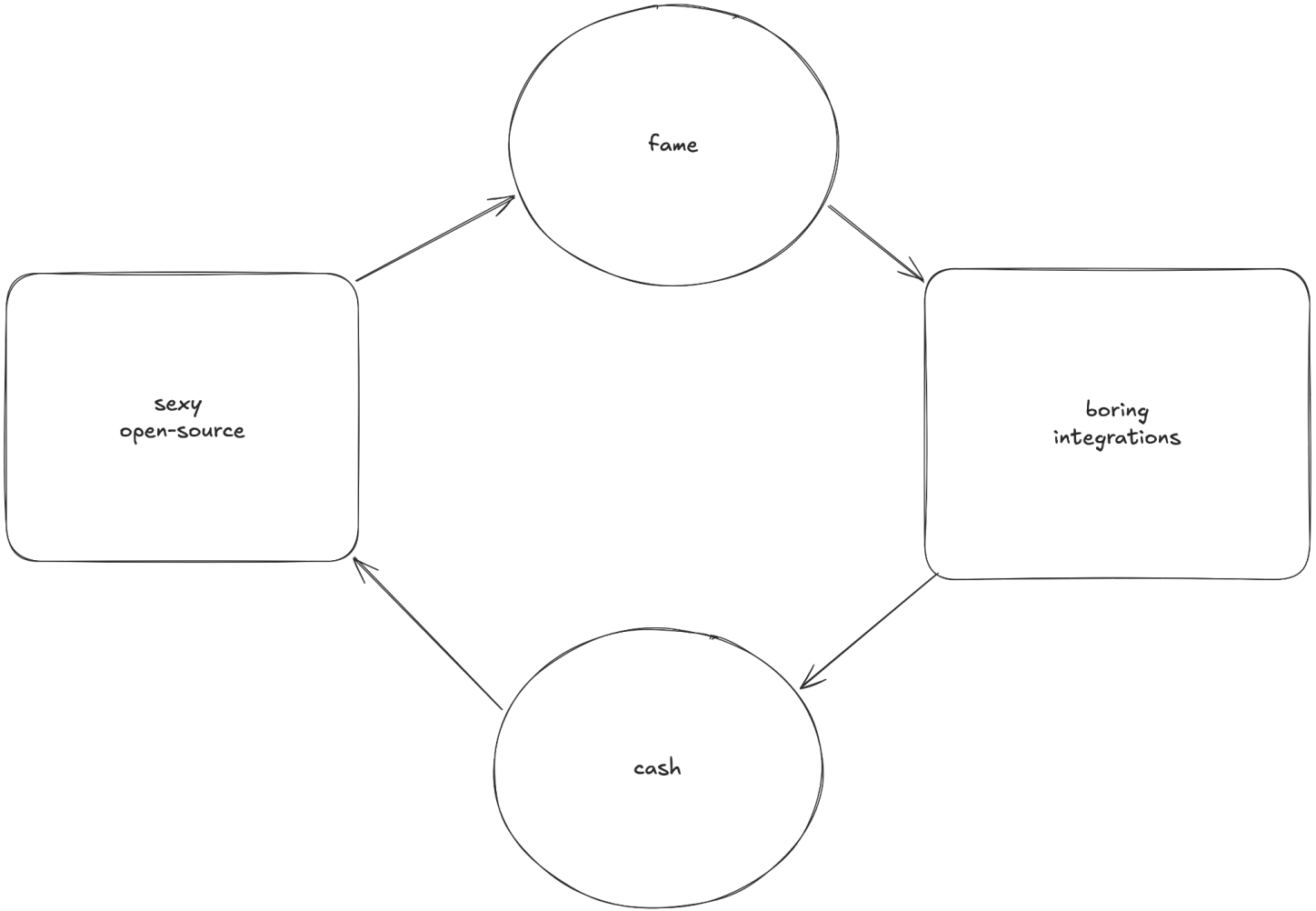
Research
Production
Strategic Open Source

- Domain specificity: DNA model, legal or financial model
- Deployment constraints: resource constraints, custom hardware like FPGA
- Safety and governance: complete control over data and model behavior

- Domain specificity: still, continuous pre-train?
- Safety and governance: does training on our data really help?

- Infrastructure for others
- Technical credibility -

- Infrastructure for others - how it helps?
- Technical credibility


fame
boring
integrations
cash
sexy
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